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The AI Ecosystem: Structure, Financing, and Financial Stability Considerations

Yingyuan Chen, Deepali Gautam, and Jason Wu

TNM/2026/13

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Abstract:
The rapid expansion of artificial intelligence (AI)-related investment has become increasingly important for economic growth, financial markets, and global financial conditions. This technical note maps the emerging AI ecosystem through an AI Value Stack and assesses its financial stability implications and balance sheet vulnerabilities. While the current cycle is anchored by highly profitable incumbent firms with strong balance sheets, investment is becoming more capital intensive and increasingly reliant on external financing. Circular financing arrangements further connect equity investment, debt financing, and future revenue commitments across firms. Combined with technological obsolescence, elevated market concentration, and rising comovement among AI-related firms, these linkages could amplify shocks if AI monetization or productivity gains disappoint. Current financial stability risks appear manageable, but vulnerabilities could build as leverage, concentration, and interconnectedness increase.

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Abbreviations

AI	artificial intelligence
Capex	capital expenditure
CPIS	Coordinated Portfolio Investment Survey
DCC-GARCH	Dynamic Conditional Correlation Generalized Autoregressive Conditional Heteroskedasticity
EBIT	earnings before interest and taxes
GPU	graphics processing unit
IIP	International Investment Position
LP	limited partnership
M-GARCH	Multivariate Generalized Autoregressive Conditional Heteroskedasticity
NBER	National Bureau of Economic Research
P/E	price-to-earnings ratio
PP&E	property, plant, and equipment
S&P 500	Standard & Poor's 500 Index
SLT	Securities Long-Term (Treasury International Capital reporting system)
TIC	Treasury International Capital System

The Macro Relevance of the AI Boom

The artificial intelligence (AI) investment boom has become macrofinancially significant. Since the public release of ChatGPT in late 2022, AI-related investment has accelerated rapidly across cloud infrastructure, semiconductors, data centers, power systems, and software deployment. Technology investment as a share of GDP has surpassed levels last observed during the dot-com period at the beginning of the new millennium (Figure 1, panel 1). Hyperscalers alone accounted for roughly $500 billion in capital expenditure (capex) in 2025, while total AI-related capex is projected to exceed $4 trillion by 2029, with close to 75 percent originating from hyperscalers.

Exceptionally strong earnings growth and the dominant market position of incumbent mega-cap firms have so far supported the current expansion, in contrast to previous technology cycles. AI-related firms have significantly outperformed the broader equity market, with the largest AI firms accounting for about 35 percent of total market capitalization and driving equity concentration measures toward historical highs. At the same time, earnings growth has remained exceptionally strong among technology firms and has supported a broader improvement in earnings growth beyond the technology sector (Figure 1, panel 2). Strong earnings have supported elevated investment and valuations, while the growing weight of AI-related firms has made the broader market increasingly sensitive to their continued earnings delivery. This trend is unlike previous technology cycles, such as the dotcom period, which did not feature companies with exceptional earnings.

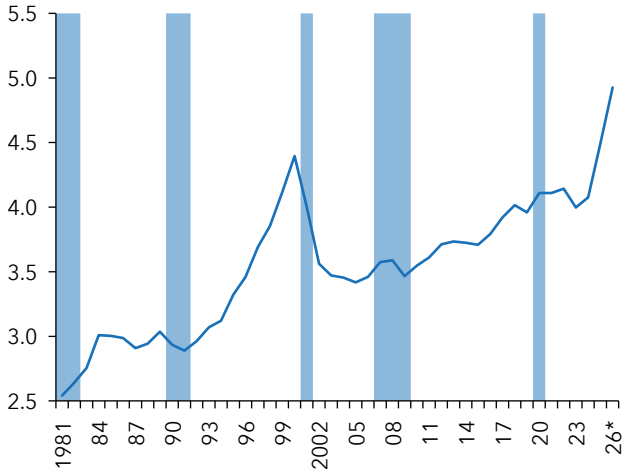
The scale of AI investment and equity market gains has in turn become an important driver of broader economic growth and financial conditions. The AI investment boom has bolstered US growth amid tighter monetary policy, tariff uncertainty, and geopolitical shocks. At the same time, the significant AI-driven gains in the stock markets have helped keep financial conditions relatively accommodative despite elevated interest rates and rising volatility elsewhere.

The AI boom is generating growing global spillover through trade, commodity, and capital-flow channels. US leadership in AI has attracted substantial foreign inflows into US markets, reinforcing the dominant role of US assets in global portfolios. Since late 2022, holdings of US equities as a share of portfolio investment assets have risen markedly across countries and jurisdictions, particularly in the euro area, the United Kingdom, and advanced Asian economies (Figure 1, panel 3). Given the growing weight of AI-related firms in US equity markets, foreign investors have become increasingly exposed to the AI investment cycle and associated market risks. At the same time, the investment cycle has generated meaningful spillovers across regions and sectors. East Asian economies have benefited through semiconductor and hardware supply chains while commodity exporters, such as in Latin America, have benefited from rising demand for copper, energy, and power infrastructure, shaping the relative market performance globally.

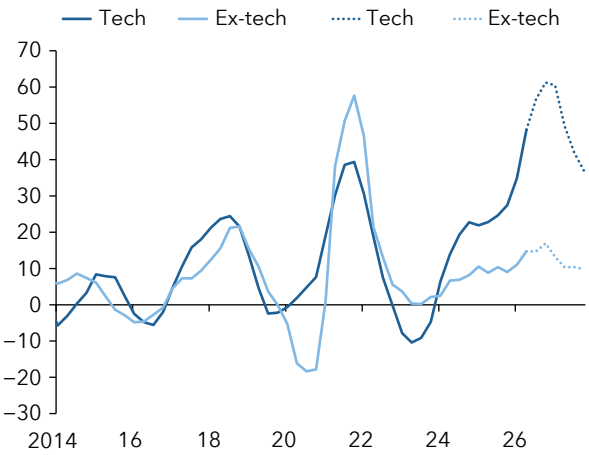
At the same time, the growing macrofinancial relevance of AI investment also raises financial stability questions. As this technical note illustrates, if investment scales further, financing needs are increasingly expanding beyond internal cash flow toward external financing and more interconnected financing structures. This shift raises the importance of assessing not only the sustainability of earnings and valuations but also the balance sheet vulnerabilities and amplification channels embedded within the broader AI ecosystem.

Figure 1. AI-Related Investment and Corporate Earnings

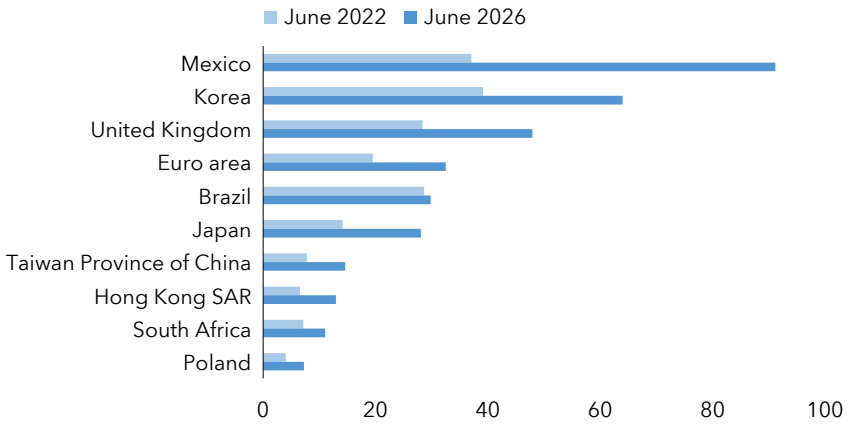
1. US Tech Investment
(Percentage of GDP)



2. S&P 500 Earnings Growth: Tech versus Ex-Tech
(Percent, year over year)



3. Holdings of US Equities across Jurisdictions
(Percentage of the country's portfolio investment assets in IIP)



Sources: Bloomberg Finance L.P.; EUROPACE AG/Haver Analytics; IMF Coordinated Portfolio Investment Survey; Treasury International Capital System; and IMF staff calculations.

Note: In panel 1, shading indicates NBER recessions, and 2026 share is the average of annualized first- and second-quarter data. Tech investment includes private nonresidential fixed investment in software and information processing equipment. In panel 3, holdings of US equities are from the Treasury International Capital's SLT table. Portfolio investment assets by residents are from IIP data. The two data sources may not capture the same sources, as TIC data are based on US securities custodian reporting, and IIP statistics are collected by local authorities AI = artificial intelligence; IIP = International Investment Position. AI = artificial intelligence; IIP = International Investment Positions; NBER = National Bureau of Economic Research; TIC = Treasury International Capital.

Mapping the AI System to Balance Sheet Vulnerabilities

AI-related firms form a broad and increasingly interconnected ecosystem that extends well beyond semiconductors and hyperscalers. Initially, this technology cycle was centered on compute and cloud platforms. It has since evolved into a broader production network spanning industrials, materials, energy, and software. This evolution reflects the compute intensity required to train and deploy frontier AI models, which demands substantial complementary investment in data centers, infrastructure, and downstream capacity.

To map the ecosystem, a stylized AI Value Stack (stack) was constructed composing five functional layers (Table 1). The stack consists of 74 firms selected to capture the largest and most economically relevant participants across the AI supply chain. While not intended to be exhaustive and currently tilted toward incumbents, the framework provides a system-level view of the AI ecosystem and a granular assessment of balance sheet vulnerabilities and interconnectedness across the stack.

Table 1. Functional Layers in the AI Value Stack

		Layer	Activities	Number of Companies	Number of Countries
Broad AI		Builders	Construction, engineering, infrastructure, industrial equipment	10	3
		Energizers	Utilities, power generation, grid, cooling, energy infrastructure	12	5
	Core AI	Chip developers	Semiconductors, GPUs, memory, networking equipment	12	4
	Core AI	Hardware developers	Device manufacturers, lithography machine manufacturers, assemblers and designers of AI server systems	9	3
	Core AI	Hyperscalers	Cloud platforms and large-scale compute providers	9	3
		Operator/ applications	Software, data managers, AI-enabled applications, services	22	6

Source: IMF staff.

Note: AI = artificial intelligence; GPUs = graphics processing units.

To assess balance sheet vulnerabilities systematically, firms are evaluated across five fundamental dimensions (Table 2). Firms are ranked cross-sectionally within each dimension, with the lowest quartile classified as weakest (red in the heatmap) and the highest quartile as strongest (green in the heatmap). Because firms span structurally different sectors, quartile thresholds provide a consistent basis for comparison while preserving heterogeneity. Thresholds are also cross-checked against S&P 500 aggregates to ensure that thresholds remain economically meaningful and not solely driven by sector composition. The final thresholds are presented in column 3 of Table 2. With these fundamental dimensions and thresholds, each layer of the AI Value Stack can be assigned a degree of vulnerability.

Table 2. Balance Sheet Vulnerability Dimensions

Fundamental Dimensions	Measures	Thresholds	
		Weakest (red)	Strongest (green)
Leverage	Debt to equity (multiples)	Over 1.5×	Below 0.5×
Liquidity	Current assets to current liabilities (multiples)	Below 1×	Over 1.5×
Profitability	Operating margin (percent)	Below 10%	Over 20%
Capex intensity	Capex to sales (multiple)	Over 35×	Below 5×
Valuation	Forward P/E (multiples)	Over 30×	Below 17.6×

Sources: Bloomberg Finance L.P.; and IMF staff.

Note: These measures are asset weighted within each layer. Capex = capital expenditure; P/E = price-to-earnings ratio.

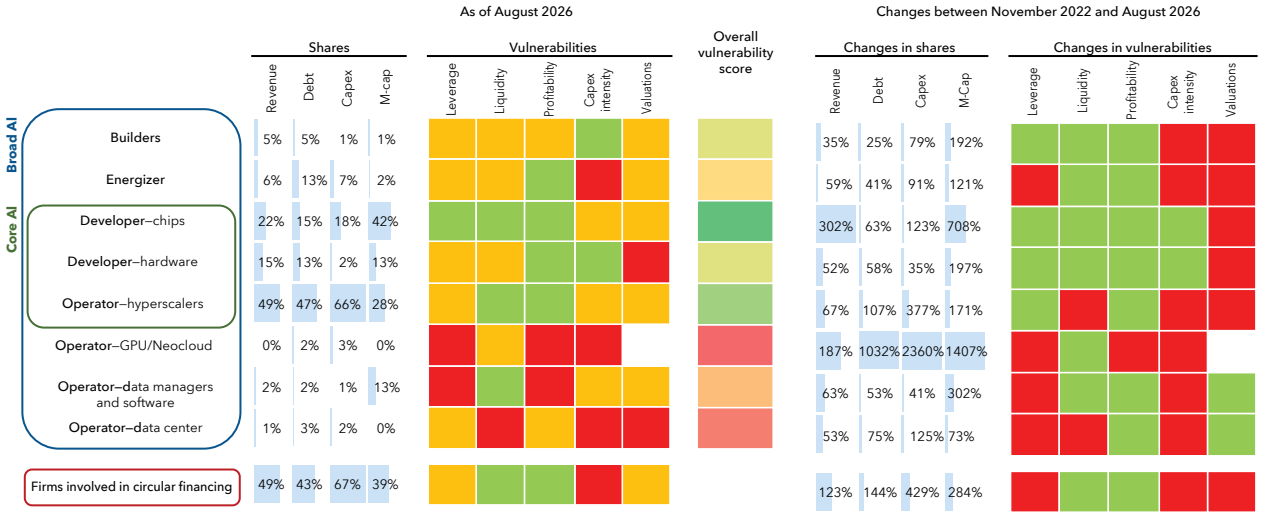
As of the second quarter of 2026, the analysis reveals substantial heterogeneity in balance sheet vulnerabilities across the stack (Figure 2).

- Core AI firms like hyperscalers and chip developers exhibit the strongest fundamentals. Profitability emerges as the primary source of resilience, with an aggregate operating margin approaching 30 percent. This margin allows these firms to sustain elevated investment without materially weakening a liquidity buffer or increasing leverage, with capex intensity remaining at moderate levels relative to other stacks.
- Downstream and infrastructure-linked layers appear materially more financially constrained. Operators and data centers rank weaker across leverage, profitability, and capital intensity. The weakest link appears to be the GPU/Neocloud—with leverage above 600 percent, still negative profitability, and capex intensity exceeding 300 percent. These firms remain heavily investment dependent and have yet to establish durable earnings profiles.
- Valuation pressures appear highest in operator and datacenter stacks. Loss-making GPU/Neocloud firms are excluded from valuation comparisons for consistency.
- Firms involved in "circular financing"—described in the next section—tend to combine strong profitability with high capital expenditure.

Strong earnings have underpinned rapid investment growth, but the balance sheet implications have increasingly diverged across the stack. Changes in vulnerability dimensions since the release of ChatGPT in November 2022 illustrate these evolving dynamics (Figure 2).

- Profitability stands out as the dimension showing the broadest improvement. Core AI stacks show particularly strong earnings growth despite a substantial increase in capex. The chip stack recorded over 300 percent revenue growth, supporting a nearly 700 percent gain in market capitalization, suggesting that valuation expansion has remained anchored by high expectations of earnings delivery.
- The investment boom has affected balance sheets differently across the stack. Among core AI firms, including hyperscalers, rapid investment growth has absorbed internal cash flow, leading to worsening liquidity and greater debt financing. However, leverage has remained contained so far. By contrast, the capital-intensive infrastructure-related stacks have increasingly relied on external financing to support expansion. This external financing has helped preserve liquidity but at the cost of higher leverage from already elevated levels. The pattern is most pronounced in the weakest stack—GPU/Neocloud—where debt has increased more than 10× alongside a 20× increase in capex.
- Valuation expansion has been broad-based despite increasing divergence in balance sheet fundamentals, underscoring investor optimism about firms across the AI ecosystem.

Figure 2. AI Value Stack and Vulnerability Assessment (as of 2026:Q2)



Sources: Bloomberg Finance L.P.; and IMF staff calculations.

Note: In the static heatmap, the color scheme follows the thresholds in Table 2. Yellow depicts values that lie between those thresholds for each measure. In the dynamic heatmap, green/red denotes improving/deteriorating relative to November 2022. The AI ecosystem includes 74 public and private firms from Australia, Canada, China, Hong Kong SAR, Ireland, Japan, Korea, The Netherlands, Switzerland, Taiwan Province of China, the United Kingdom, and the United States. More specifically, companies in the sample are Aecom, Jacobs Solutions Inc, Fluor Corp, Emcor Group Inc, Comfort Systems USA Inc, Duke Energy Corp, Entergy Corp, Xcel Energy Inc, Sempra, Alliant Energy Corp, Vertiv Holdings, Eaton Corp Plc, Nvent Electric Plc, Nvidia Corp, Advanced Micro Devices, Intel Corp, Broadcom Inc, Marvell Technology Inc, Amkor Technology Inc, Super Micro Computer Inc, Celestica Inc, Jabil Inc, Amazon.com Inc, Microsoft Corp, Alphabet Inc, Oracle Corp, Coreweave Inc, Nebius Group N.V., Equinix Inc, Digital Realty Trust Inc, Iron Mountain Inc, GDS Holdings Ltd, Meta Platforms Inc, Palantir Technologies Inc, Taiwan Semiconductor, Samsung Electronics Co Ltd, ASML Holding N.V., Graphcore Ltd, Apple Inc, Qualcomm Inc, Mediatek Inc, SK Hynix Inc, Micron Technology Inc, Cisco Systems Inc, Arista Networks Inc, Juniper Networks Inc, Schindler Holding AG-Reg, Abb Ltd-Reg, Generac Holdings Inc, Caterpillar Inc, Cummins Inc, Mitsubishi Electric Corp, Delta Electronics Inc, Johnson Controls Internation, Daikin Industries Ltd, Alibaba Group Holding Ltd, NTT Data Group Corp, Global Data Centre Group, Cyrusone Holdco LLC, QTS Realty Trust Inc, Coresite Realty Corp, Snowflake Inc, Informatica Inc, Scale AI Inc, VMware Inc, Canon Inc, Confluent Inc, Coherent Corp, Intl Business Machines Corp, Tencent Holdings Ltd, Baidu Inc, Cerebras Systems Inc, Iren Ltd, and Applied Digital Corp. AI = artificial intelligence; Capex = capital expenditure; GPUs = graphics processing units; M-cap = market capitalization.

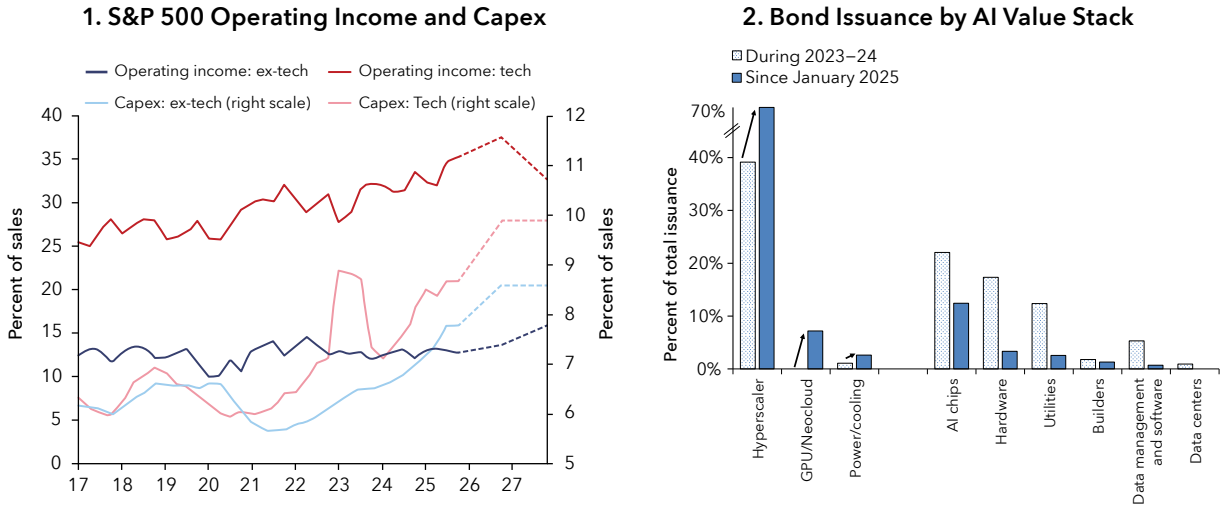
Some of the most systemically relevant nodes remain private and operate with limited disclosure. Foundational model labs—including OpenAI, Anthropic, and other frontier infrastructure and application firms—have become central nodes despite remaining outside public markets. The three largest private foundational model labs are collectively valued at over $2 trillion based on their latest funding rounds, despite generating less than $75 billion of combined revenue as of August 2026. Anticipation of future public listings has also attracted significant retail interest through publicly traded vehicles and other structures that provide indirect pre-IPO exposure to private AI firms (DXYZ ETF, Hyperliquid).

The rapid valuation gains among private AI firms have also supported the earnings of some strategic investors, including hyperscalers. Revaluation of these stakes can generate substantial gains for investors, raising reported earnings and lowering valuation measures; however, because these gains do not reflect recurring operating income, they raise questions about future earnings quality. In addition, these exposures create an additional link between private AI valuations and the balance sheets of public firms, potentially amplifying the impact of a broader repricing across the AI ecosystem.

Financing the AI Boom

The scale of the current AI investment boom has shifted financing from internal cash flows toward external sources. Historically, large technology firms funded most investment through operating cash flows, supported by high margins, asset-light business models, and strong balance sheets (Figure 3, panel 1). While profitability remains exceptionally strong, capital intensity has risen sharply as firms increase spending on data centers, compute infrastructure, power systems, and associated facilities. At the same time, many of the largest firms continue to return substantial capital to shareholders through dividends and buybacks. As a result, the projected scale of AI-related investment increasingly exceeds what can be comfortably financed internally. While large hyperscalers remain financially strong, external financing is expected to play a growing role as investment requirements continue to expand. In the absence of additional financing, firms would increasingly face the trade-off between sustaining investment plans and shareholder distributions.

Figure 3. Capex and Financing



Sources: Bloomberg Finance L.P.; and IMF staff calculations.

Note: Panel 2 shows the shares of the $530 billion in bonds issued by the firms in the AI stack shown in Figure 2 since 2023, of which $405 billion has been issued since 2025. AI = artificial intelligence; Capex = capital expenditure.

Financing sources are also becoming more diversified. Traditional corporate bonds remain an important channel, with over $530 billion raised by AI-related firms included in the stack since 2023, dominated by core AI and GPU/Neocloud firms (Figure 3, panel 2).[1] In addition, the growing funding needs have supported the expansion of alternative financing structures. Estimates suggest that approximately $2.9 trillion of datacenter capital expenditure through 2028 will be financed through a combination of hyperscaler cash flow, bilateral credit, private capital, public debt, and securitization markets (April 2026 *Global Financial Stability Report*).

[1] Firms included in the AI Value Stack represent 53 percent of all artificial intelligence–related bond issuances in Canada, the United States, Ireland, The Netherlands, Switzerland, the United Kingdom, Australia, China, Hong Kong SAR, Japan, South Korea, and Taiwan Province of China since 2023, and 65 percent since January 2025.

Private credit has emerged as one of the fastest-growing financing channels. The flexibility of private credit allows lenders to fund large, specialized projects that may not fit traditional bank lending or public debt markets, including datacenter construction, equipment financing, and asset-backed lending structures. Although system-wide exposures remain relatively modest (Aldasoro and others 2026), the expansion of private credit increasingly links commercial banks, nonbank lenders, and AI-related borrowers, embedding the capitalization of this technological expansion within a more complex, interconnected credit ecosystem.

Investor demand for such financing has remained strong so far. Despite a sharp increase in projected borrowing needs, AI-related issuers have generally retained favorable market access and have not faced a significant spread penalty relative to similarly rated issuers. However, current financing conditions are based not only on expectations of continued earnings delivery but also on assumptions regarding asset durability, collateral values, and the resilience of increasingly complex financing arrangements. As investment scales further and financing structures become more interconnected, vulnerabilities may emerge even in the absence of a material slowdown in AI demand.

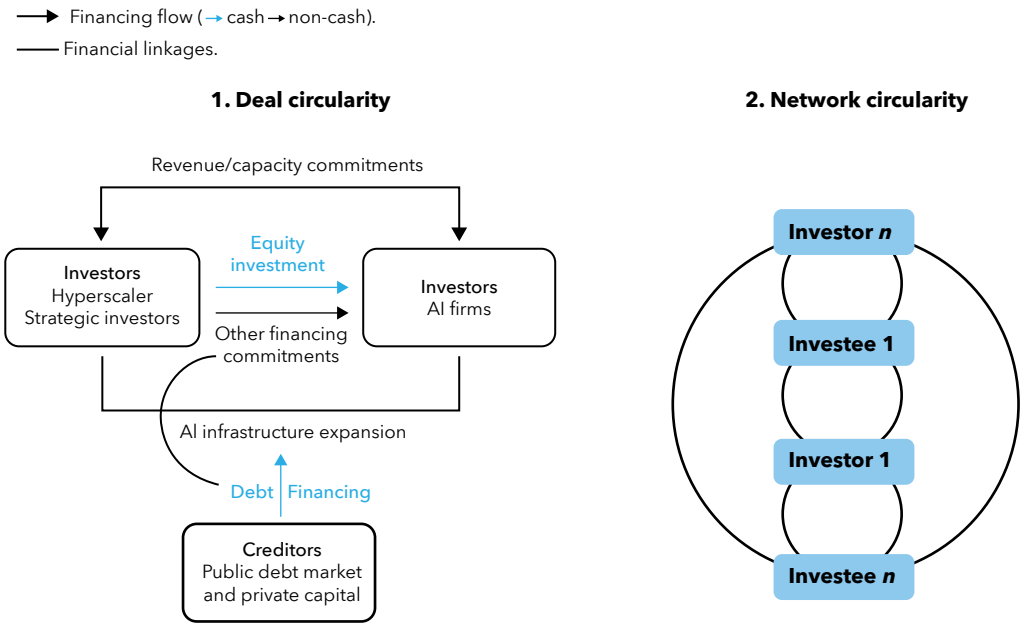
Circular Financing Structures Increase Opacity, Concentration, and Amplification Risk

Circular financing increasingly links investment, financing, and commercial relationships across the AI ecosystem. Unlike traditional supply-chain financing, which mainly supports working capital, large firms in the AI stack finance suppliers, developers, and infrastructure providers while simultaneously acting as customers of those same firms, a structure known as "circular financing." In many cases, investors' equity investment is accompanied by revenue commitments that effectively tie the investees' future operating performance back to the original investors and other financing commitments, such as long-term leases or project financing guarantees. These commercial and financing commitments support infrastructure expansion while facilitating additional debt financing, allowing relatively modest equity investments to underpin substantially larger financing structures. These arrangements can also extend beyond a bilateral relationship, with other AI firms, financial institutions, and private investors participating in the same financing arrangements.

These overlapping relationships create both deal- and network-level circularity, increasing financial interconnectedness across the AI ecosystem. The integration of investment, financing, and commercial relationships within a single structure creates deal circularity, which establishes multiple channels of financial dependence among the counterparties. These structures are then replicated across a broad set of firms, creating network circularity wherein investors and investees form overlapping relationships with multiple counterparties. As a result, public and private firms with diverse business models and balance sheet vulnerabilities become increasingly interconnected (Figure 4).

The publicly disclosed circular financing arrangements already involve economically significant firms and substantial financial commitments. The IMF staff have investigated 14 of the largest circular financing deals in the United States that have public disclosures and were subject to media reports. Publicly listed firms that are involved in these deals are captured by the AI stack (Figure 2) and account for close to 50 and 40 percent of revenue and market capitalization, respectively. When publicly listed firms are combined with private AI-related firms involved in the deals, their market-implied valuation is about $20 trillion. Current estimates on financing and revenue commitments likely represent a lower bound on the eventual scale of a circular financing arrangement. Nonetheless, focusing on these 14 deals with circularity features reveal some notable features.

Figure 4. Circular Financing Schematic



Source: IMF staff.
Note: AI = artificial intelligence.

Direct equity exposures appear relatively modest for investors but support substantially larger revenue and financing commitments. Total direct equity investment across the deals amounts to approximately $300 billion, equivalent to about 1.4 percent of the investors' aggregate market capitalization. These transactions are also associated with future revenue commitments totaling around $1.4 trillion over an average horizon of five years, implying an annual gross return on equity investment of approximately 65 percent for this period (Figure 5, panel 1). While the direct exposures appear manageable for investors on a deal-by-deal basis, they likely understate the total exposures as other financial commitments—off-balance-sheet commitments, financing guarantees, and other financing arrangements—could be embedded in these deals.

The leverage embedded in the arrangements appears large at the investee level, where equity investment has been followed by substantial debt financing. Out of 14 deals, 9 were followed by additional debt financing totaling over $650 billion—implying aggregate deal-level leverage of about 130 percent—with the largest transactions also exhibiting the highest leverage (Figure 5, panel 2). In addition, these equity financings appear to have been raised at a substantial discount to prevailing market valuations. Based on the associated revenue commitments, the implied deal-level price-to-sales multiple averages around 1.5×, less than one-tenth of the multiple implied by the investee's own public market valuations. At the same time, these revenue commitments may constitute a substantial contingent burden given that many investees remain in expansion mode and have yet to establish sustained cash-flow generation.

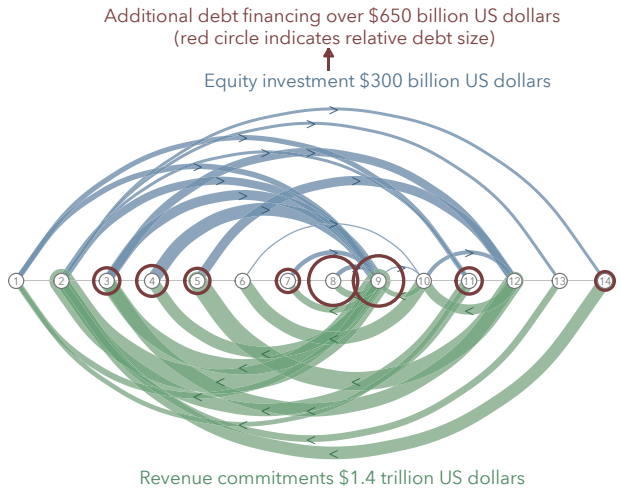
Presently, the central role of large firms largely mitigates risks with strong balance sheets involved in these circular deals, while vulnerabilities are more concentrated among investees. High profitability and cash buffers allow large firms to sustain elevated capex without immediate solvency or liquidity concerns, while

debt markets continue to absorb issuance. By contrast, many investees face rising debt and off-balance-sheet commitments that could become increasingly difficult to sustain if earnings delivery fall short of expectations.

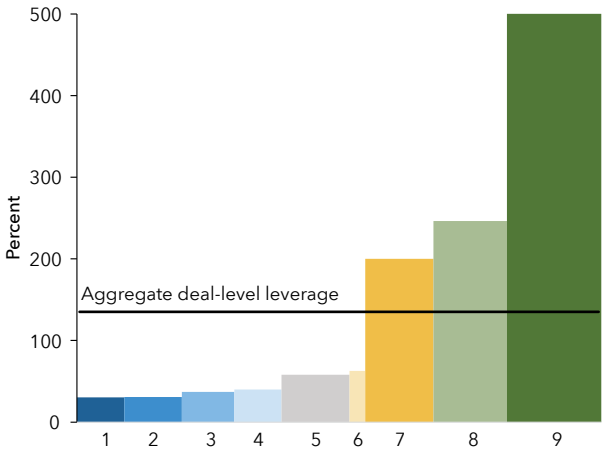
Circular financing could nevertheless amplify shocks by transmitting stress across the network. Strong earnings and successful monetization can reinforce valuations, financing availability, and investment expansion. Conversely, weaker-than-expected monetization could simultaneously reduce revenues, impair collateral values, and tighten financing conditions. Stress or failure of key investees could then propagate through the circular financing network, impairing expected revenue streams for investors and other counterparties and further weakening earnings and financing capacity across the network. These interconnected relationships could therefore enhance the initial shock and ultimately weigh on future investment.

Figure 5. Circular Financing

1. Circular Financing: Equity, Debt, and Revenue Commitments



2. Deal-Level Leverage for Investees: Total Debt Raised to Initial Equity Investment



Sources: Bloomberg Finance L.P.; media reports; and IMF staff estimates.

Note: In panel 1, each number represents a firm engaged in AI circular financing. Numbered circles denote firms; 1 through 10 refer to investors, and 9 through 14 refer to investees. Some firms are both investors and investees. Blue and green bands indicate the size of equity investments and future revenue commitments, respectively. Red circles indicate debts issued as part of the circular deals, including debt backed by investor guarantees, contingent financing, and other off-balance-sheet debt. In panel 2, the width of the bars indicates the relative size of combined market capitalization of investor and investee.

A key source of uncertainty is where the associated financial exposures ultimately reside. Public information provides only a partial view of who holds AI-related debt. Off-balance-sheet commitments further complicate the assessment of ultimate exposures. This opacity makes it difficult to assess the potential for direct losses that could propagate through the financial system. Even if direct exposures of systemically important financial institutions remain contained, a sharp retrenchment in AI investment could trigger broader macro-financial spillover. Such a pullback would manifest itself through weaker investment and growth, declining equity valuations and household wealth, and tightening financial conditions.

Financial Stability Risks of Weaker-than-Expected AI Economics: Amplification Channels

As the AI investment cycle scales, the shift toward external financing and a more interconnected financing structure could increase sensitivity of the ecosystem to shocks. Shorter effective asset lives (obsolescence risk) and rising market fragility from correlated positioning across AI-related firms could further amplify these vulnerabilities.

Obsolescence Risk May Increase Financing Needs

AI investment is increasingly concentrated in assets whose economic lifespans may be materially shorter than their financing horizons. Based on reported depreciation schedules, financial disclosures from major hyperscalers indicate an estimated average useful lifespan of approximately seven years for property, plant, and equipment. However, GPUs and advanced chips—which account for a substantial share of AI infrastructure costs—may become obsolete substantially faster given rapid technological progress (Figure 6, panel 1).

Faster obsolescence could weaken internal funding capacity and increase financing needs as firms are required to replace assets more frequently. Shorter useful lives affect balance sheets through two mechanisms. First, accelerated depreciation reduces earnings and internal funding capacity. Second, firms must reinvest more frequently to maintain productive capacity, increasing reliance on debt financing. Scenario analysis in the April 2026 *Global Financial Stability Report* suggests that reducing useful life assumptions from seven years to three years lowers aggregate earnings before interest and taxes margins by more than 9 percentage points. Under a high capital-intensity scenario, operating profitability can be fully absorbed by replacement investment (Figure 6, panel 2). Under a stylized debt-financing assumption (see Figure 6 note), aggregate hyperscaler debt rises from approximately $800 billion to more than $1 trillion, while modeled credit spreads widen by roughly 60 basis points, compared with current spreads that range between 20 and 160 basis points.

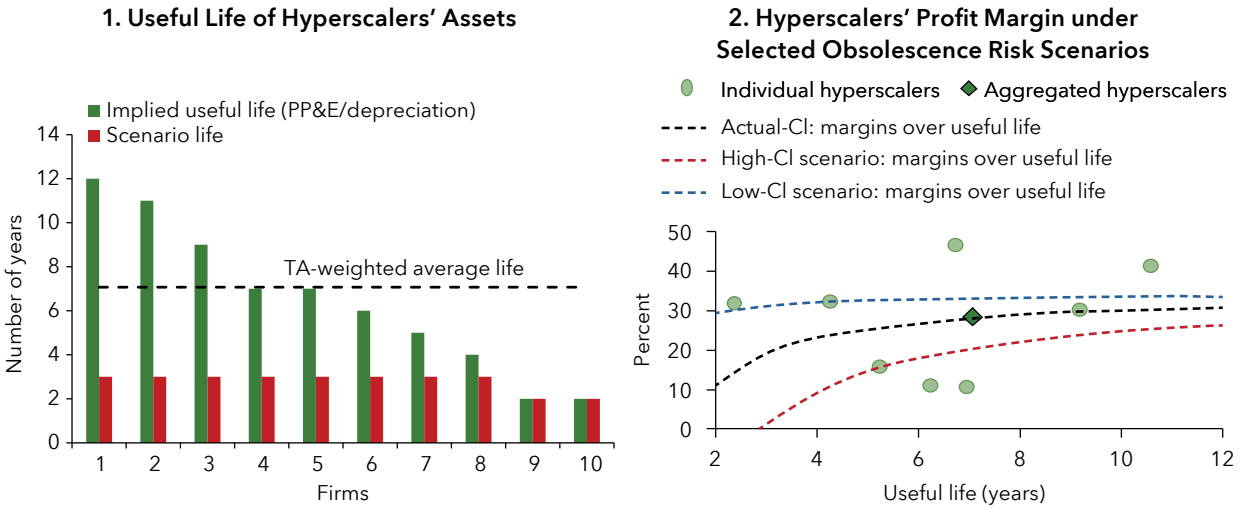
These effects remain primarily a business risk rather than an immediate financial stability concern but will become increasingly relevant as investment scales. Importantly, these mechanisms could occur even if demand for AI services and AI-related investments continues—faster-than-expected obsolescence could raise financing pressures through accelerated replacement investment and refinancing risks for debt backed by these assets.

Rising Correlated Positioning Raises Market Fragility

AI-related stocks have become the dominant driver of US equity market performance. Out of the approximately $36 trillion increase in S&P 500 market capitalization since late 2022, about $21 trillion has come from firms in the AI stacks. The AI cycle has therefore become increasingly important not only for sectoral performance but also for broader index returns, household wealth, and global portfolio allocation.

AI-related stocks are exhibiting increasing co movement, with returns increasingly moving as a common group rather than as differentiated firms. Conditional return correlations estimated using DCC-GARCH

Figure 6. Obsolescence Risk



Sources: Bloomberg Finance L.P.; London Stock Exchange Group Workspace; and US Securities and Exchange Commission filings.

Note: In panel 1, useful lives are proxied by dividing PP&E by depreciation based on information from individual US Securities and Exchange Commission 10K filings or other company disclosures. In panel 2, the light-green circles indicate the current EBIT margin of individual hyperscalers as of the end of 2025. Diamond marker indicates the aggregated EBIT margin. Dotted lines indicate aggregated EBIT margins under different useful lives, applying a widely used straight line depreciation methodology (see Online Annex 1.6 of the April 2026 *Global Financial Stability Report*). The "low-CI scenario" hypothetically assumes a case with fixed assets amounting to 10 percent of HSs' aggregated revenue, whereas the "high-CI scenario" assumes fixed assets of 100 percent. Aggregated HSs' CI, the property-and-equipment-to-revenue ratio, is 46 percent. CI = capital intensity; EBIT = earnings before interest and taxes; HS = hyperscaler; PP&E = property, plant, and equipment; TA = total asset.

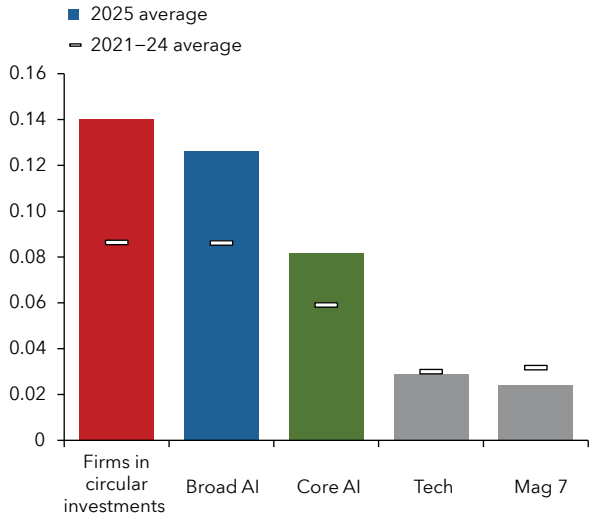
models have risen markedly and now exceed those observed among the broader technology sector and the "Magnificent 7" stocks. The increase in co-movement is particularly pronounced among firms involved in circular financing—consistent with increasingly intertwined financing and commercial relationships—followed by core AI firms, which are at the center of the investment and financing cycle (Figure 7, panel 1).

Elevated correlations may amplify downside risks even without a broad deterioration in macroeconomic or earnings fundamentals. Counterfactual exercises using M-GARCH imply cumulative returns could decline by approximately 8–10 percent under stressed correlation assumptions relative to baseline conditions (Figure 7, panel 2).

The amplification effect may become more pronounced when combined with elevated concentration and interconnected financing structures. In this environment, a reassessment of AI monetization or investment returns could trigger simultaneous repricing across highly correlated firms, tightening financing conditions and amplifying market volatility through deleveraging and portfolio rebalancing channels.

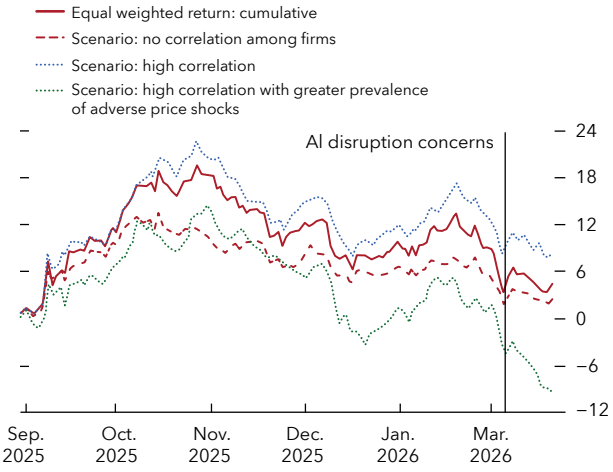
Figure 7. Market Fragility from Interconnectedness

1. Conditional Correlation of AI-Related Stock Returns



2. Cumulative Return since September 2025 for Select Firms Engaged in Circular Financing

(Percent)



Sources: Bloomberg Finance L.P.; and IMF staff estimates.

Note: Panel 1 grouping corresponds to AI Value Stack. Panel 1 is based on the bivariate DCC-GARCH model using weekly returns of stock prices since 2016. In panel 2, equity returns for hyperscalers operating specifically within the circular financing structure are jointly modeled as a multivariate-GARCH system allowing for dynamic variance–covariance matrices, assuming disturbances obey Student's *t*-distribution. Return data used are at a daily frequency and the estimation sample starts on January 1, 2020. Returns modeled in the system are first adjusted to purge influence of the broad market (systematic) risk factor by orthogonalizing relative to daily returns on the S&P 500. Cumulative returns under different scenarios represent an average across 50,000 model simulations for the period September 1, 2025, to March 31, 2026. Mag 7 = Magnificent 7 (Alphabet, Amazon, Apple, Meta Platforms, Microsoft, Nvidia, and Tesla).

Financial Stability Outlook and Policy Considerations

The current AI boom differs in key respects from prior technology cycles. Unlike the late 1990s, the current expansion is led by incumbent firms with substantial earnings, cash flows, and financing capacity, allowing investment to expand without a commensurate rise in leverage. Valuations are elevated but remain below dotcom extremes, while much of the increase in market capitalization has been accompanied by strong earnings growth rather than purely speculative multiple expansion.

However, concentration risks appear materially larger. AI-related firms account for a historically large share of equity market capitalization and index performance. The overall size of financial markets relative to the economy has also risen substantially, supported in part by increasing foreign ownership of US equities. As a result, the macrofinancial footprint of the AI ecosystem extends well beyond the technology sector itself. A correction would likely affect not only equity valuations but also investment, household wealth, private credit, and cross-border capital flows. Consequently, even a correction considerably smaller than the dotcom period could generate significant macrofinancial spillovers.

The sustainability of the AI boom ultimately depends on whether productivity gains and earnings growth materialize at a pace sufficient to justify current investment plans.

- On the upside, AI adoption could generate substantial productivity gains, support earnings growth, and strengthen economic resilience. Investment in data centers, power infrastructure, and related supply chains could broaden growth beyond the technology sector. At the same time, successful deployment of AI applications could support profitability across a wider range of sectors. In this scenario, strong earnings growth would validate current investment spending and help sustain both financing conditions and valuations.
- The downside scenario could emerge through either a sudden interruption in investment or a gradual deterioration in AI economics.
    - An exogenous shock—a recession, funding disruption, geopolitical escalation, or tighter regulation—could trigger an abrupt slowdown in capital expenditures.
    - More gradually, monetization of AI services may prove weaker than expected, leading firms to scale back investment expansion.
    - A related risk is that AI ultimately falls short of the transformational productivity gains currently embedded in market expectations. In this scenario, demand for compute capacity and associated infrastructure may continue to grow, but at a materially slower pace than anticipated.
    - Once spending slows, pressures would likely propagate through the broader ecosystem, initially affecting firms that must rely on continued investment expansion or external financing before spilling over to the broader economy through investment, employment, and asset price channels.
    - Financing structures could amplify these dynamics. As previously indicated, obsolescence risks could increase refinancing pressures, while circular financing arrangements may amplify contagion across firms, investors, and funding channels.

Consequently, the key financial stability question is whether the scale, concentration, and interconnectedness of the AI ecosystem could enable a sectoral shock to generate broader global macrofinancial spillovers. While detailed policy recommendations are outside the scope of this note, measures aimed at improving transparency, monitoring interconnected exposures, and identifying leverage-related vulnerabilities may become increasingly important as AI-related financing expands and financial linkages become more complex. Policy frameworks should also remain attentive to the accumulation of AI-related systemic risks across the financial system, including excessive leverage and risk-taking among bank and nonbank financial institutions. (IMF 2026).

More broadly, the rapidly evolving AI ecosystem highlights the need for financial stability frameworks that remain forward looking and adaptable to new sources of risk. Enhanced data collection on AI-related exposures, scenario-based risk assessments, and monitoring of cross-border spillovers can strengthen the ability to identify emerging vulnerabilities and potential amplification channels before they become systemic (Barhoumi and others 2026).

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